Two-stage Gauss–Legendre Method

Exact one-step Jacobian, symplecticity, and symmetry

Fourth-order collocation method for canonical Hamiltonian systems

1 Hamiltonian convention

Let $z \in \mathbb{R}^{2d}$ and consider the possibly time-dependent canonical Hamiltonian system

$$\dot{z} = f(t,z) = \Omega^{-1}\nabla_z H(t,z), \qquad \Omega = \begin{pmatrix} 0 & I_d \\ -I_d & 0 \end{pmatrix}. \tag{1}$$

For the component-major guiding-centre state

$$z = (x_1, \dots, x_d, y_1, \dots, y_d)^T,$$

this convention gives

$$f(t,z) = \begin{pmatrix} -\nabla_y H(t,z) \\ \nabla_x H(t,z) \end{pmatrix}.$$

At any fixed time and state, let $F = D_z f(t,z)$. Since $\nabla_z^2 H$ is symmetric,

$$F^T\Omega + \Omega F = 0. \tag{2}$$

This identity is the infinitesimal Hamiltonian condition used below. The proof remains valid for a time-dependent Hamiltonian because the stage times do not depend on the initial state z_n.

2 Gauss–Legendre stages

Set

$$r = \frac{\sqrt{3}}{6}, \qquad c_1 = \frac{1}{2} - r, \qquad c_2 = \frac{1}{2} + r.$$

The Butcher coefficients of the two-stage Gauss–Legendre method are

$$A = \begin{pmatrix} \frac{1}{4} & \frac{1}{4} - r \\ \frac{1}{4} + r & \frac{1}{4} \end{pmatrix}, \qquad b = \begin{pmatrix} \frac{1}{2} \\ \frac{1}{2} \end{pmatrix}. \tag{3}$$

For one step of length h, the stage states satisfy

$$Z_i = z_n + h\sum_{j=1}^{2} a_{ij} f(t_n + c_j h, Z_j), \qquad i = 1,2, \tag{4}$$

and the accepted state is

$$z_{n+1} = \Phi_{h,t_n}(z_n) = z_n + h\sum_{i=1}^{2} b_i f(t_n + c_i h, Z_i). \tag{5}$$

The method is implicit: equations (4) determine (Z_1, Z_2) simultaneously.

3 Direct proof of fourth-order accuracy

We now prove directly why the Runge–Kutta order conditions through degree four give a fourth-order method. For clarity, first consider the autonomous problem

$$\dot{y} = f(y), \qquad y(t_n) = y. \tag{6}$$

The non-autonomous case follows by augmenting the state with time, $\widehat{y} = (t, y)$ and $\dot{\widehat{y}} = (1, f(t, y))$. Because $c = A\mathbf{1}$, applying the Runge–Kutta method to this augmented autonomous system produces exactly the stage times used in (4).

Set

$$F = f(y), \qquad F' = Df(y), \qquad F'' = D^2 f(y), \qquad F''' = D^3 f(y), \tag{7}$$

where the derivatives are multilinear maps. Repeated application of the chain rule gives

$$\begin{aligned} y' &= F, \\ y'' &= F'F, \\ y''' &= F''(F,F) + F'^2 F, \\ y^{(4)} &= F'''(F,F,F) + 3F''(F,F'F) + F'F''(F,F) + F'^3 F. \end{aligned} \tag{8}$$

Consequently, the exact flow has the expansion

$$\begin{aligned} y(t_n + h) = y + hF &+ \frac{h^2}{2} F'F \\ &+ \frac{h^3}{6}\left[F''(F,F) + F'^2 F\right] \\ &+ \frac{h^4}{24}\left[F'''(F,F,F) + 3F''(F,F'F) + F'F''(F,F) + F'^3 F\right] + O(h^5). \end{aligned} \tag{9}$$

For a generic Runge–Kutta step, write

$$K_i = f(Y_i), \qquad Y_i = y + h\sum_j a_{ij} K_j, \qquad y_{n+1} = y + h\sum_i b_i K_i, \tag{10}$$

and define $C = \mathrm{diag}(c)$. Powers such as c^2 and c^3 are understood componentwise. If $\Delta_i = h\sum_j a_{ij} K_j$, Taylor expansion about y gives

$$K_i = F + F'\Delta_i + \frac{1}{2}F''(\Delta_i, \Delta_i) + \frac{1}{6}F'''(\Delta_i, \Delta_i, \Delta_i) + O(h^4). \tag{11}$$

Substituting the stage expansions recursively into (11) and collecting equal powers of h yields

$$\begin{aligned} K_i = F &+ hc_i F'F \\ &+ h^2\left[(Ac)_i F'^2 F + \frac{1}{2}c_i^2 F''(F,F)\right] \\ &+ h^3\left[(A^2 c)_i F'^3 F + \frac{1}{2}(Ac^2)_i F'F''(F,F)\right. \\ &\qquad \left. + c_i (Ac)_i F''(F,F'F) + \frac{1}{6}c_i^3 F'''(F,F,F)\right] + O(h^4). \end{aligned} \tag{12}$$

Inserting (12) into the accepted update in (10) gives

$$\begin{aligned} y_{n+1} = y &+ h(b^T\mathbf{1})F + h^2(b^T c)F'F \\ &+ h^3\left[(b^T Ac)F'^2 F + \frac{1}{2}(b^T c^2)F''(F,F)\right] \end{aligned}$$

$$+ h^4 \Big[(b^T A^2 c) F'^3 F + \frac{1}{2}(b^T A c^2) F' F''(F,F) + (b^T C A c) F''(F, F'F) + \frac{1}{6}(b^T c^3) F'''(F,F,F) \Big] + O(h^5). \tag{13}$$

The elementary differentials in (9) and (13) are independent for a general smooth vector field. Their coefficients agree through h^4 precisely when

$$\begin{aligned} b^T \mathbf{1} &= 1, & b^T c &= \frac{1}{2}, \\ b^T c^2 &= \frac{1}{3}, & b^T A c &= \frac{1}{6}, \\ b^T c^3 &= \frac{1}{4}, & b^T C A c &= \frac{1}{8}, \\ b^T A c^2 &= \frac{1}{12}, & b^T A^2 c &= \frac{1}{24}. \end{aligned} \tag{14}$$

For example, the coefficient of $F''(F,F)$ in the numerical step is $\frac{1}{2} b^T c^2$ and its exact coefficient is $\frac{1}{6}$, which gives $b^T c^2 = \frac{1}{3}$. The other seven equalities follow from the same term-by-term comparison. Thus these conditions are not merely formal identities: they are exactly the equations that make the numerical and exact Taylor expansions coincide through fourth degree.

It remains to verify them for the coefficients in (3). Direct calculation gives

$$b^T \mathbf{1} = 1, \qquad b^T c = \frac{1}{2}, \qquad b^T c^2 = \frac{1}{3}, \qquad b^T c^3 = \frac{1}{4}, \tag{15}$$

and

$$Ac = \frac{1}{2} c^2, \qquad b^T A = b^T (I - C). \tag{16}$$

Therefore

$$\begin{aligned} b^T A c &= \frac{1}{2} b^T c^2 = \frac{1}{6}, & b^T C A c &= \frac{1}{2} b^T c^3 = \frac{1}{8}, \\ b^T A c^2 &= b^T (c^2 - c^3) = \frac{1}{12}, & b^T A^2 c &= \frac{1}{2} b^T A c^2 = \frac{1}{24}. \end{aligned} \tag{17}$$

Hence all conditions in (14) hold, and comparison of (9) with (13) proves the one-step defect estimate

$$\| y(t_n + h) - y_{n+1} \| = O(h^5). \tag{18}$$

Finally, let e_n be the global error. For a smooth Lipschitz vector field, the Runge–Kutta one-step map is Lipschitz with factor $1 + Lh$ for sufficiently small h. The local estimate (18) then gives

$$\| e_{n+1} \| \le (1 + Lh) \| e_n \| + C h^5. \tag{19}$$

For $nh \le T$, the discrete Gronwall estimate yields

$$\| e_n \| \le C h^5 \sum_{j=0}^{n-1} (1 + Lh)^j \le C_T h^4. \tag{20}$$

Thus the two-stage Gauss–Legendre method has global order four.

4 Exact Jacobian of the stage root

Introduce the stage field Jacobians and stage sensitivities

$$F_i := D_z f(t_n + c_i h, Z_i), \qquad S_i := \frac{\partial Z_i}{\partial z_n}. \tag{21}$$

Differentiating (4) with respect to z_n gives

$$S_i = I_{2d} + h \sum_{j=1}^{2} a_{ij} F_j S_j. \tag{22}$$

Equivalently,

$$\underbrace{\begin{pmatrix} I_{2d} - \frac{h}{4} F_1 & -h\left(\frac{1}{4} - r\right) F_2 \\ -h\left(\frac{1}{4} + r\right) F_1 & I_{2d} - \frac{h}{4} F_2 \end{pmatrix}}_{M = D_{(Z_1, Z_2)} R} \begin{pmatrix} S_1 \\ S_2 \end{pmatrix} = \begin{pmatrix} I_{2d} \\ I_{2d} \end{pmatrix}. \tag{23}$$

Here M is the Jacobian of the coupled stage residual. It is also the matrix used by an exact Newton linearization, but it is *not* the Jacobian of the physical one-step map.

Assuming that the converged stage root is regular, the implicit function theorem gives

$$\begin{pmatrix} S_1 \\ S_2 \end{pmatrix} = M^{-1} \begin{pmatrix} I_{2d} \\ I_{2d} \end{pmatrix}. \tag{24}$$

In an implementation, equation (23) should be solved directly; the inverse in (24) is only notation.

5 Exact Jacobian of the physical step

Define the Jacobian of the map (5) by

$$\mathcal{J}_n := D_{z_n} \Phi_{h, t_n}(z_n).$$

Differentiating the accepted update and using (21) yields

$$\boxed{\mathcal{J}_n = I_{2d} + h \sum_{i=1}^{2} b_i F_i S_i = I_{2d} + \frac{h}{2} \left(F_1 S_1 + F_2 S_2\right).} \tag{25}$$

Combining (24) and (25), the same Jacobian can be written as

$$\boxed{\mathcal{J}_n = I_{2d} + h \begin{pmatrix} \frac{1}{2} F_1 & \frac{1}{2} F_2 \end{pmatrix} M^{-1} \begin{pmatrix} I_{2d} \\ I_{2d} \end{pmatrix}.} \tag{26}$$

This is the exact tangent of the ideal converged root. It differentiates the mathematical Gauss–Legendre map, not a finite Newton stopping rule.

6 Direct proof of symplecticity with the step Jacobian

The one-step map is symplectic if and only if

$$\mathcal{J}_n^T \Omega \mathcal{J}_n = \Omega. \tag{27}$$

Set

$$K_i := F_i S_i. \tag{28}$$

Then equations (25) and (22) become

$$\mathcal{J}_n = I_{2d} + h \sum_i b_i K_i, \qquad I_{2d} = S_i - h \sum_j a_{ij} K_j. \tag{29}$$

Expanding the symplecticity defect gives

$$\begin{aligned} \mathcal{J}_n^T \Omega \mathcal{J}_n - \Omega &= h \sum_i b_i \left(K_i^T \Omega + \Omega K_i \right) \\ &\quad + h^2 \sum_{i,j} b_i b_j K_i^T \Omega K_j. \end{aligned} \tag{30}$$

By the Hamiltonian identity (2),

$$\begin{aligned} K_i^T \Omega S_i + S_i^T \Omega K_i &= S_i^T \left(F_i^T \Omega + \Omega F_i \right) S_i \\ &= 0. \end{aligned} \tag{31}$$

Using the second identity in (29) on both occurrences of I_{2d} therefore gives

$$K_i^T \Omega + \Omega K_i = -h \sum_j a_{ij} \left(K_i^T \Omega K_j + K_j^T \Omega K_i \right). \tag{32}$$

Substitute (32) into (30). Renaming i and j in the second term arising from (32) produces

$$\boxed{\mathcal{J}_n^T \Omega \mathcal{J}_n - \Omega = h^2 \sum_{i,j} \left(b_i b_j - b_i a_{ij} - b_j a_{ji} \right) K_i^T \Omega K_j.} \tag{33}$$

Consequently, the defect vanishes whenever the Runge–Kutta coefficients obey

$$\boxed{b_i a_{ij} + b_j a_{ji} = b_i b_j, \qquad i, j = 1, 2.} \tag{34}$$

7 Verification for the Gauss–Legendre coefficients

For the diagonal terms in (34),

$$2b_1a_{11} - b_1^2 = 2\left(\frac{1}{2}\right)\left(\frac{1}{4}\right) - \frac{1}{4} = 0, \qquad 2b_2a_{22} - b_2^2 = 0. \tag{35}$$

For the off-diagonal terms,

$$\begin{aligned} b_1a_{12} + b_2a_{21} - b_1b_2 &= \frac{1}{2}\left(\frac{1}{4} - r\right) + \frac{1}{2}\left(\frac{1}{4} + r\right) - \frac{1}{4} \\ &= 0. \end{aligned} \tag{36}$$

The case $(i, j) = (2, 1)$ is the same equality with the two summands exchanged. Thus every coefficient multiplying $K_i^T \Omega K_j$ in (33) is zero, and hence

$$\boxed{\mathcal{J}_n^T \Omega \mathcal{J}_n = \Omega.} \tag{37}$$

Therefore the two-stage Gauss–Legendre method is symplectic when its implicit stage equations are solved exactly.

8 Symmetry and time reversibility

Symmetry is a property distinct from symplecticity. The method is symmetric when its backward step is exactly the inverse of its forward step. For the possibly time-dependent problem (1), this means

$$\boxed{\Phi_{-h,t_n+h} \circ \Phi_{h,t_n} = \mathrm{Id}\,.} \tag{38}$$

For an autonomous problem, equation (38) reduces to $\Phi_{-h} = \Phi_h^{-1}$.

For the forward step, write

$$f_1 = f(t_n + c_1 h, Z_1), \qquad f_2 = f(t_n + c_2 h, Z_2). \tag{39}$$

Equations (4) and (5) are then

$$\begin{aligned} Z_1 &= z_n + h(a_{11} f_1 + a_{12} f_2), \\ Z_2 &= z_n + h(a_{21} f_1 + a_{22} f_2), \qquad (40) \\ z_{n+1} &= z_n + h(b_1 f_1 + b_2 f_2). \qquad (41) \end{aligned}$$

Now start from z_{n+1} at time $t_n + h$ and apply the same method with step $-h$. Its stages $\widehat{Z}_1, \widehat{Z}_2$ satisfy

$$\widehat{Z}_i = z_{n+1} - h \sum_{j=1}^{2} a_{ij} f(t_n + h - c_j h, \widehat{Z}_j). \tag{42}$$

The Gauss nodes are reflected about the midpoint:

$$1 - c_1 = c_2, \qquad 1 - c_2 = c_1. \tag{43}$$

Thus the backward stage times are the forward stage times in reverse order. We shall verify that the backward stages are

$$\boxed{\widehat{Z}_1 = Z_2, \qquad \widehat{Z}_2 = Z_1.} \tag{44}$$

Under the identification (44), the first backward stage equation becomes

$$\begin{aligned} \widehat{Z}_1 &= z_{n+1} - h(a_{11} f_2 + a_{12} f_1) \\ &= z_n + h\left[(b_1 - a_{12}) f_1 + (b_2 - a_{11}) f_2\right]. \end{aligned} \tag{45}$$

For the coefficients (3),

$$b_1 - a_{12} = a_{21}, \qquad b_2 - a_{11} = a_{22}. \tag{46}$$

Consequently,

$$\widehat{Z}_1 = z_n + h(a_{21} f_1 + a_{22} f_2) = Z_2.$$

Likewise, the second backward stage satisfies

$$\begin{aligned} \widehat{Z}_2 &= z_{n+1} - h(a_{21} f_2 + a_{22} f_1) \\ &= z_n + h\left[(b_1 - a_{22}) f_1 + (b_2 - a_{21}) f_2\right]. \end{aligned} \tag{47}$$

Since

$$b_1 - a_{22} = a_{11}, \qquad b_2 - a_{21} = a_{12}, \tag{48}$$

we obtain

$$\widehat{Z}_2 = z_n + h(a_{11} f_1 + a_{12} f_2) = Z_1.$$

The proposed reversed stages therefore solve the coupled backward stage equations exactly.

The output of the backward step is now

$$\begin{aligned}\widehat{z}_n &= z_{n+1} - h(b_1 f_2 + b_2 f_1) \\ &= z_n + h(b_1 f_1 + b_2 f_2) - h(b_1 f_2 + b_2 f_1) \\ &= z_n, \end{aligned} \tag{49}$$

where the last equality uses $b_1 = b_2 = 1/2$. This proves (38).

More generally, an s-stage Runge–Kutta method is symmetric when its coefficients are invariant under stage reversal:

$$\boxed{c_i = 1 - c_{s+1-i}, \qquad b_i = b_{s+1-i}, \qquad a_{ij} = b_{s+1-j} - a_{s+1-i,s+1-j}.} \tag{50}$$

The coefficients in (3) satisfy all three identities for $s = 2$.

Differentiating the map identity (38) also gives the Jacobian form of reversibility:

$$\boxed{D\Phi_{-h,t_n+h}(z_{n+1})\, D\Phi_{h,t_n}(z_n) = I_{2d}.} \tag{51}$$

Thus the backward-step Jacobian is exactly the inverse of the forward-step Jacobian. As with symplecticity, a finite Newton tolerance and floating-point round-off can produce a small measured reversibility defect even though the ideal implicit method is exactly symmetric.

9 Implemented configurations

The public `GaussLegendre4` model uses the same collocation equations for every configuration. The selector `newton_jacobian_method` accepts `analytic`, `finite_difference`, or `auto`. Analytic mode uses exact guiding-centre particle blocks, finite-difference mode differentiates the coupled stage residual, and auto resolves the available dynamics capability. Absolute and relative tolerances and the maximum iteration count control the finite root solve.

With `track_energy=True`, the method also advances the time-conjugate momentum using the two converged Gauss stages and monitors the generalized Hamiltonian $K = H + k$. This triangular extension does not alter the physical collocation stages. These are configurations of one Gauss–Legendre model, not separate numerical methods.

10 Consequences and numerical interpretation

For one guiding-centre particle, $\mathcal{J}_n \in \mathbb{R}^{2\times 2}$ and

$$\mathcal{J}_n^T \Omega \mathcal{J}_n = \det(\mathcal{J}_n)\Omega.$$

Equation (37) therefore implies

$$\det(\mathcal{J}_n) = 1,$$

so the one-step map preserves oriented area. In dimensions greater than two, unit determinant is only a consequence of symplecticity, not an equivalent test.

If $\mathcal{J}_n$ denotes the local Jacobian at step n, the accumulated flow tangent after N steps is

$$\mathcal{P}_N = \mathcal{J}_{N-1} \cdots \mathcal{J}_1 \mathcal{J}_0.$$

Since a product of symplectic matrices is symplectic,

$$\mathcal{P}_N^T \Omega \mathcal{P}_N = \Omega.$$

The algebraic result concerns the exact root and the exact tangent (25). In floating-point computations, a finite Newton tolerance, round-off, or a finite-difference approximation of the map Jacobian produces a small nonzero measured defect

$$\left\| \mathcal{J}_n^T \Omega \mathcal{J}_n - \Omega \right\|.$$

That numerical floor does not contradict the exact symplecticity proved above.

References

- J. M. Sanz-Serna, “Runge–Kutta schemes for Hamiltonian systems,” *BIT Numerical Mathematics*, 28 (1988), 877–883.
- E. Hairer, C. Lubich, and G. Wanner, *Geometric Numerical Integration*, second edition, Springer, 2006.

11 Physical accepted states and passive energy tracking

For N particles with d physical components, every accepted snapshot contains $Z \in \mathbb{R}^{dN}$, one scalar time t, and optional separate physical momentum $\kappa \in \mathbb{R}^N$. Thus GC accepted vectors have $2N$ components and FC vectors have $4N$, whether energy is tracked or not. A doubled method prepares temporary $(u, v) \in \mathbb{R}^{2dN}$ copies, allows them to separate, and projects to one physical result before acceptance. The representation stores only the physical size and never owns a clock, momentum, physical coordinate declaration, dynamics or solver.

The normalized passive balance is

$$\dot{\kappa} = -\partial_t H(t, Z), \qquad \kappa(t_0) = 0, \qquad \Delta K(t) = H(t, Z(t)) + \kappa(t) - H(t_0, Z_0).$$

GC split-map contributions are summed in temporary work and divided by two to obtain physical momentum. Tracking leaves physical stages, nonlinear stopping scales and adaptive acceptance unchanged. An autonomous Hamiltonian supplies an explicit zero momentum derivative.

Hairer's constraint is solely $u - v = 0$: the reduced multiplier has $2N$ entries and ABBA's simultaneous spatial solve has $6N$ unknowns. Neither t nor κ is projected. The former eight-component projection is retired. Dynamics declare component order and position/velocity roles through immutable structural data; initial conditions and results retain those data independently. Methods own maps, projection equations and momentum updates. Generic copy operations live in `src/formulations/state.py`.

Controllers and compiled JAX loops carry physical values and momentum separately. Observers receive physical states. Energy histories in `States.diagnostics` retain their existing labels; `extended_time` has shape (N, n_{samples}) and is broadcast from scalar sample times. Fixed shadow samples leave accepted states and counters unchanged. Adaptive quadrature uses accepted dense output and physical-only error control.

The two converged Gauss stage states supply the passive momentum quadrature.

12 Optional JAX execution

An optional JAX driver executes the same numerical map on one selected CPU or GPU, with particle operations batched inside a compiled sequential time loop. It reuses the canonical coefficients and prepared potential splines. Physical-state tolerance scaling, signed stage times, passive energy weights and independent shortened output maps retain their definitions above. Applicable Newton systems use independent particle blocks; an explicitly selected Broyden iteration retains its global secant matrix. Nonlinear iteration limits and failure checks remain active on the device.

Both execution routes use double precision and return the same physical solution layout. Floating-point reassociation can produce round-off differences without changing the designed order or geometric properties. First-call compilation is separate from warmed integration timings, which include transfers and output construction. The sibling runtime document and shared execution contract specify supported callbacks and exact device scope.